

**BACHELOR OF COMPUTER
APPLICATIONS [BCA (REVISED)]**

Term-End Examination

June, 2025

**BCS-041 : FUNDAMENTALS OF COMPUTER
NETWORKS**

Time : 3 Hours

Maximum Marks : 100

Note : *Question No. 1 compulsory. Attempt any
three questions from the rest. Use of
calculator is allowed.*

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1. (a) Discuss the need of modulation in computer network. Also, differentiate between the Amplitude and Frequency Modulation. 5

- (b) What is time division multiplexing ?
Give the advantages and disadvantages
of time division multiplexing. 5
- (c) Why network models are divided into
layers ? Write the similarities between
TCP/IP and OSI model. 5
- (d) Explain the working of RARP (Reverse
Address Resolution Protocol) with the
help of a suitable diagram. 5
- (e) How is CRC (Cyclic Redundancy
Check) code used for error detection in
digital network ? Give an example to
illustrate your answer. 5
- (f) Differentiate between "Peer-to-Peer"
and "Client-Server" networking. 5

- (g) Explain silly window syndrome at transport layer, with suitable example. 5
- (h) Discuss the sublayers of data link layer. Also, give characteristics of these sublayers. 5
2. (a) Write the steps of RSA algorithm. Assume two primary members $p = 3$ and $q = 11$, use RSA algorithm to calculate encryption and decryption keys. 10
- (b) Write the working of Selective Repeat method. Also, compare it with Go-Back-N using example. 10

3. (a) Explain the functions of the following network devices. Also, give *one* advantage and *one* disadvantage of each : 10
- (i) Hub
 - (ii) Bridge
 - (iii) Repeater
 - (iv) Modem
 - (v) Switch
- (b) What is SNMP ? Discuss the functions performed by SNMP for network management. 10
4. (a) Write Link state routing algorithm. Explain its working with the help of an example. 10
- (b) Compare 1G, 2G and 3G wireless generations, based on the following criteria : 10
- (i) Communication method

(ii) Modulation technique

(iii) Services

(iv) Channel Assignment

5. Write short notes on the following : $5 \times 4 = 20$

(i) ICMP

(ii) ALOHA

(iii) Sliding Window Protocol

(iv) TCP/IP Model

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Term-End Examination

December, 2025

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Maximum Marks : 100

Note : *Question No. 1 compulsory. Attempt any
three questions from the rest. Use of
calculator is allowed.*

1. (a) How does synchronous transmission differ from asynchronous transmission ?
Give suitable example for each. 5

- (b) What is frequency division multiplexing ? Give the advantages and disadvantages of frequency division multiplexing. 5
- (c) What are gateways ? Explain the importance of gateways in networking. 5
- (d) Explain the working of ARP (Address Resolution Protocol) with the help of a suitable diagram. 5
- (e) Justify the statement, “Slotted ALOHA achieves double efficiency than pure ALOHA.” 5
- (f) Differentiate between ‘Public-key’ and ‘Private-key’ cryptography. 5

- (g) Explain count to infinity problem with the help of an example. 5
- (h) Define angle modulation. Give its types. Also, discuss the limitation of angle modulation. 5
2. (a) Assume message $M = 1010101010$ bits and generator $G = 10001$ bits. Explain, how CRC (Cyclic Redundancy Check) is used for error detection using the given message bits and generator bits. 10
- (b) Discuss distance vector routing using a subnet topology. Also, discuss the different metrics used in distance vector routing algorithm. 10
3. (a) What is MD5 ? Write and explain the steps of MD5 algorithm. 10

- (b) Write the role of data link layer in OSI model. Explain the services and functions provided by data link layer. Also, give the importance of its sub-layers. 10
4. (a) Explain X.25 architecture with the help of a suitable diagram. How does X.25 architecture differ from frame relay ? Give the advantages of frame relay over X.25 architecture. 10
- (b) Differentiate between IPv4 and IPv6. Also, discuss the need of IPv6. Find the classes of the following IPv4 addresses : 10
- (i) 193.14.56.22
 - (ii) 226.11.14.27
 - (iii) 134.11.27.13
 - (iv) 252.5.15.111

5. Write short notes on the following : $4 \times 5 = 20$

- (i) Leaky Bucket Algorithm
- (ii) Multimode Optical Fiber
- (iii) OSI Model
- (iv) CSMA/CD

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**BACHELOR OF COMPUTER
APPLICATIONS (B. C. A.) (REVISED)**

Term-End Examination

June, 2024

**BCS-041 : FUNDAMENTALS OF
COMPUTER NETWORKS**

Time : 3 Hours

Maximum Marks : 100

Note : (i) *Question no. 1 is compulsory. Attempt any **three** questions from the rest.*

(ii) *Use of calculator is allowed.*

1. (a) Compare parallel and serial communication. Why is serial data transmission faster than parallel data transmission ? Explain. 5

(b) Compare DHCP and SNMP. Give importance of DHCP and SNMP for the application layer of TCP/IP model. 5

- (c) Discuss the role of switch as an inter-networking device. Compare layer-2 switch with layer-3 switch. 5
- (d) What is multiplexing ? Briefly discuss the importance of multiplexing. List the basic multiplexing techniques. 5
- (e) What is the function of Datalink layer ? Briefly discuss the role of two sub-layers of Datalink layer. 5
- (f) What is distance vector routing ? Briefly discuss the problem of distance vectors routing. 5
- (g) What is stream cipher ? Give *two* advantages and *two* disadvantages of stream cipher. 5
- (h) Differentiate between Symmetric and Asymmetric cryptography. 5
2. (a) Discuss the term 'Cyclic Redundancy Check (CRC)'. Find CRC, if the message is $X^7 + X^5 + 1$ and the generator polynomial is $X^3 + 1$. 10

- (b) Differentiate between pure ALOHA and slotted ALOHA. If the throughput of pure ALOHA is $S = G e^{-2G}$, show that the maximum throughput ($S_{\max}$) is 0.184. 10
3. (a) What is classful addressing ? How is it different from classless addressing ? How does classless addressing result in the decrease of table size ? Given the network address 17.0.0.0, find the class, the block and the range of address. 10
- (b) Explain RSA algorithm with suitable example and block diagram. 10
4. (a) Differentiate between following : 5+5
- (i) ICMP and IGMP
- (ii) Token bucket algorithm and leaky bucket algorithm.
- (b) Briefly discuss the functions of various layers involved in TCP/IP model. Also mention the protocols defined under each layer. 10

5. Write short notes on the following : $4 \times 5 = 20$

- (a) MD5 Digest
- (b) Components of address field in the Frame Relay Protocol Data Unit
- (c) GSM Architecture
- (d) Frame Relay
- (e) 3-way handshake protocol

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Note : *Question number 1 is compulsory. Attempt any **three** questions from the rest. Use of calculator is allowed.*

1. (a) Differentiate between private key and public key cryptography. Give suitable example of each. 6
- (b) Briefly discuss the concept of frequency shift keying and phase shift keying. Give an application of each. 6

- (c) What is count to infinity problem in distance vector routing protocol ? How does it happen ? Explain with an example. 10
- (d) Briefly discuss the functions of Layer-2 switch and Layer-3 switch. Compare Switch with Hub. 6
- (e) How does pure ALOHA differ from slotted ALOHA ? Explain. 6
- (f) What are virtual circuits ? Discuss the effect of router failure in virtual circuits. 6
2. (a) Differentiate between ARP and RARP. Explain the working of ARP using a diagram. 10
- (b) How does classful addressing differ from classless addressing ? How does classless addressing result in decrease in the table size ? 10
3. (a) Explain the concept *Go-Back-N sliding window protocol* with a suitable example and diagram. 10

- (b) What is Fragmentation ? Explain why IPv4 and IPv6 protocols need to fragment some packets. 10
4. (a) Write the step-by-step working of link state routing. Also, compare it with distance vector routing. 10
- (b) Discuss the concept of sliding window protocol with the help of an example. Also, explain how piggybacking technique works. 10
5. (a) What is IGMP ? Draw the header fields of IGMP. Also, explain the significance of each field. 10
- (b) Write the significance and usage of the following networking devices : 2×5=10
- (i) Repeater
 - (ii) Bridges
 - (iii) Switches
 - (iv) Gateways
 - (v) Networks interface card

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Note : *Question No. 1 is compulsory. Answer any
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calculator is allowed.*

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1. (a) What is a Frame Relay ? Explain the advantages of Frame Relay over X.25 Network.

6

- (b) How are Hubs, Switches and Routers different from each other ? 6
- (c) Explain POP and IMAP. What are the advantages of IMAP over POP ? 6
- (d) Compare CSMA/CD and Ethernet protocol. 6
- (e) How does circuit switching and packet switching differ ? Give merits and demerits of both. 6
- (f) Write the steps of Distance Vector Routing Algorithm. Show case the working of this algorithm with an example. 10
2. (a) Explain the working of 3-way handshake used in TCP using a suitable diagram. 10
- (b) What is Windowing ? How are flow control and reliability achieved through windowing at transport layer ? 10

3. (a) List various connecting devices in a LAN.

Explain the functioning of each. Also show the interconnectivity between devices with suitable diagram. 10

- (b) Briefly discuss the utility of CRC.

Calculate CRC if the message is $x^7 + x^5 + 1$ and the generator polynomial is $x^3 + 1$. 10

4. (a) Differentiate between Analog and Digital

Modulation. Compare and contrast between ASK, PSK and FSK (digital modulation techniques). 10

- (b) What is MD5 ? Write step by step procedure for generating 128 bit MD5 digest. 10

5. (a) Differentiate between adaptive and non-adaptive routing. Explain the working of hierarchical routing using suitable topological structure and routing table. 10
- (b) Discuss the error control techniques used at the data link layer. Also write the step by step procedure of Go-BACK-N ARQ Method. 10

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Maximum Marks : 100

Note : Question no. 1 is **compulsory**. Attempt any **three** questions from the rest. Use of calculator is allowed.

1. (a) Briefly discuss the term CRC. Determine the CRC for the data polynomial $X^5 + X^4 + X^2 + 1$ with generator polynomial $X^3 + 1$. 7
- (b) Explain how routing and switching is done in ATM network. 10
- (c) Differentiate between Classful addressing and Classless addressing. Explain how classless addressing results in decrease in table size. 10

- (d) What is a Congestion Control Mechanism ? Discuss slow start phase of TCP congestion control mechanism through an illustration. 7
- (e) What do 10Base-T and 100Base-T stand for ? Also, differentiate between the two. 6
- 2.** (a) Compare Ring topology and Mesh topology, in terms of the following parameters : 6
- (i) Requirement of cable
 - (ii) Reliability
 - (iii) Performance of network by adding extra nodes
- (b) What is Count-to-Infinity problem ? Explain. 4
- (c) Explain the multiplicative decrease process, with respect to congestion control. 4
- (d) What is Silly Window Syndrome ? What are the proposed solutions to this syndrome ? 6
- 3.** (a) Answer the following :
- (i) Given the network address 125.0.0.0, find the class, the block and the range of the address. 3
 - (ii) How can we prove that we have 2, 147, 483, 648 addresses in class A ? 2
- (b) Discuss the pros and cons of a wireless communication system. 5

- (c) How does MD5 message digest algorithm work ? Explain. 5
- (d) Explain Client-Server model of network, with the help of a diagram. 5
4. (a) Discuss the parity bit method for error detection ? Suppose a bit sequence 110001010111 is received. Assume odd parity bit method is used. Find whether received bit sequence is correct or not. 10
- (b) Explain X.25 architecture, with the help of a diagram. 10
5. Write short notes on the following : $5 \times 4 = 20$
- (a) Count-to-Infinity Problem in Distance Vector Routing Protocol
- (b) Quality of Service (QoS) of Network
- (c) Functions of DHCP
- (d) Functions of SNMP
- (e) CSMA/CD
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Note : Question no. 1 is **compulsory**. Attempt any **three** questions from the rest. Use of calculator is allowed.

1. (a) What is MD5 Digest ? Explain the steps for the process of generating 128 bit MD5 digest from any given number and key. 10
- (b) How is sampling done from analog signals ? Explain through an illustration. 7
- (c) Given data frame as 1101011011 and generator polynomial $G(x) = x^4 + x + 1$, derive the transmitted frame using CRC method. Write all the steps involved in the process. 10

- (d) How many networks can each IP address class A, B and C have ? Also find the number of hosts per network in each of the given address class. 6
- (e) Discuss the working of ARP and RARP. Also, differentiate between the two. 7
- 2.** (a) Discuss the following in context of network management : 6
- (i) Configuration management
 - (ii) Accounting management
- (b) What are the important features of an ATM switch ? 4
- (c) What are the two categories of ICMP messages ? Give two examples of each. 6
- (d) Differentiate between Virtual circuit and Datagram. 4
- 3.** (a) Explain the following features of IPv6 : 5
- (i) Tunneling
 - (ii) Dual IP Stack
- (b) What is the problem with PSK ? Explain how it may be solved. 5
- (c) What is TCP's sliding window ? Explain Silly Window Syndrome with the help of a diagram. 10

4. (a) Explain Time Division Multiplexing (TDM). Discuss the advantages and disadvantages of TDM. Also, give applications of TDM. 10
- (b) What is Three-way handshaking method ? Draw and explain the connection establishment and termination in TCP using Three-way handshaking method. 10
5. (a) Differentiate between the following : 10
- (i) Circuit switching and Packet switching
- (ii) IPv4 and IPv6
- (b) Write short notes on the following : 10
- (i) RSA
- (ii) OSI Model
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**BACHELOR OF COMPUTER APPLICATIONS
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Term-End Examination

June, 2021

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Time : 3 hours

Maximum Marks : 100

Note : Question no. 1 is **compulsory**. Attempt any **three** questions from the rest. Use of calculator is allowed.

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1. (a) Which type of communication, digital or analog, is better for computers ? Justify your answer. 5
 - (b) What are 'Hash functions' ? Why are they called 'One-way functions' ? Explain. 5
 - (c) Compare SVC and PVC of X.25 architecture. 5
 - (d) Why is serial data transmission faster than parallel data transmission ? Explain. 5

- (e) How are the number of twists in the UTP cable related to its performance ? 5
- (f) Name the two sublayers of Data Link Layer, and explain the characteristics of these layers. 5
- (g) Explain Silly Window Syndrome, with the help of a diagram. 5
- (h) Briefly discuss the problem with PSK and explain its possible solution. 5
- 2.** (a) Compare and contrast the amplitude modulation and phase modulation techniques. Discuss the limitations of each. 10
- (b) Explain RSA algorithm with example. 10
- 3.** (a) Explain the working of 3-way handshake used in TCP, using a suitable diagram. 10
- (b) Discuss the ATM service classes and QoS (Quality of Services) parameters of each class. 10
- 4.** (a) Calculate the CRC for a 10 bit sequence 1010011110 with a divisor of 1011.
Note : Show all steps and calculations. 10
- (b) Compare Time Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM). Give the advantage and disadvantage (if any) of both TDM and FDM. 10

5. Write short notes on the following :

4×5=20

- (a) CSMA/CD
 - (b) IEEE 802.3
 - (c) Frame Relay
 - (d) OSI Model
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